

CHAPTER 1

INTRODUCTION

A beam is a structural element designed to carry loads perpendicular to its longitude. Beams play a crucial role in many engineering projects, from bridges and buildings to machines and mechanical systems. These structures often face dynamic forces that make them vibrate. Knowing how a beam responds to vibration is vital, since too much movement can cause damage, make structures unsafe, or lower their efficiency. For these reasons, engineers carefully study beam vibrations during the design process.

A common and straightforward way to understand beams is by using the Euler–Bernoulli beam theory. This model works best for beams that are much longer than they are thick, and assumes that the cross-section stays flat when the beam bends. It ignores effects like shear deformation and rotary inertia, which keeps things simple and accurate for slender beams. Typically, this approach is reliable when the beam’s length is at least ten times its depth.

In this project, the free vibration behaviour of a uniform Euler–Bernoulli beam is studied. The beam is supported by a pin at the left end and a vertical sliding support at the right end. The pinned end does not allow any displacement but allows rotation, while the sliding end allows vertical movement but restricts horizontal displacement and rotation. This combination of boundary conditions makes the problem slightly different from standard cases and affects both the natural frequencies and mode shapes of the beam.

The approach followed in this project begins by deriving the governing differential equation for the beam's transverse vibration using basic principles of mechanics. The boundary conditions are then applied, and the equation is simplified using non-dimensionalization. After that, the method of separation of variables is used to obtain the general solution. From this, the frequency equation is derived, and the first few natural frequencies and corresponding mode shapes are calculated. The mode shapes are then plotted using MATLAB to better understand the vibration behaviour.

Finally, a finite element model of the beam is created in ANSYS for modal analysis. The natural frequencies and mode shapes obtained from the simulation are compared with the analytical results, and any differences are discussed.

The analysis is carried out under the assumptions that the beam is uniform, linearly elastic, and undergoes small deflections. Effects such as damping and shear deformation are neglected in accordance with Euler–Bernoulli beam theory.

CHAPTER 2

LITERATURE REVIEW

The Euler-Bernoulli beam theory is the standard framework for studying the free vibration of slender, uniform beams. This theory is widely used because it simplifies the governing equations by ignoring shear deformation and rotary inertia. It assumes that plane cross-sections stay perpendicular to the beam's axis while bending (Rao, 2010). Although this one-dimensional approach gives accurate engineering estimates for long, thin structures, Reddy (1993) points out that these simplifications can lead to small errors in higher vibration modes. In these cases, the beam's physical thickness and complex internal effects become more important.

A beam's vibration characteristics are mainly determined by its boundary conditions. In a pinned-sliding (guided) setup, the pinned end prevents movement but allows rotation; the sliding end permits vertical movement while keeping rotation at zero. Chepkwony and Ratnam (2024) show that these specific constraints can greatly change the natural frequencies and establish the allowable "wavelengths" of the mode shapes. Generally, more restrictive boundary conditions raise the system's overall stiffness, resulting in higher natural frequencies. This trend needs careful consideration when transitioning from standard supports to non-standard guided supports (Rao, 2010).

Modern engineering often confirms these analytical theories with Finite Element Method (FEM) simulations in software like ANSYS. Adavadkar and Admuthe (2021) found that modal analysis in ANSYS usually matches Euler-Bernoulli predictions within 1%. This close agreement often makes the two methods nearly the same for simple shapes. Such consistency supports using FEM as a link between ideal mathematical models and real working conditions, especially where complex shapes make exact manual calculations challenging.

However, small differences between analytical math and computer simulations can still occur. These "minute errors" often arise from converting a smooth, continuous beam into a series of discrete "blocks" or elements (Reddy, 1993). Aspects like mesh refinement, numerical round-off, and the software's handling of boundary constraints contribute to these discrepancies. While FEM is very reliable, its accuracy depends on how well the discretisation process is done, particularly in higher modes where the numerical solution needs to approach the exact analytical value.

CHAPTER 3

PROBLEM SPECIFICATION AND FORMULATION

3.1 PROBLEM SPECIFICATION

- A uniform Euler- Bernoulli beam of length L is considered.
- The beam has flexural rigidity EI , mass density ρ , and cross-sectional area A .
- The left end of the beam is supported by a pin support.
- The right end is constrained by a vertical sliding support.
- The sliding support allows vertical displacement but restricts horizontal movement and rotation.

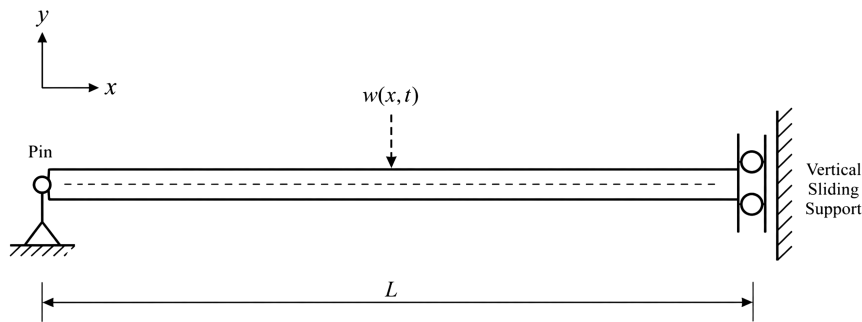


Fig 3.1 Beam with Pinned-Slider boundary conditions

3.1.1 Assumptions

- small deflection
- beam is slender ($L/h > 10$)
- linear elastic
- no damping
- Euler–Bernoulli theory (no shear deformation)

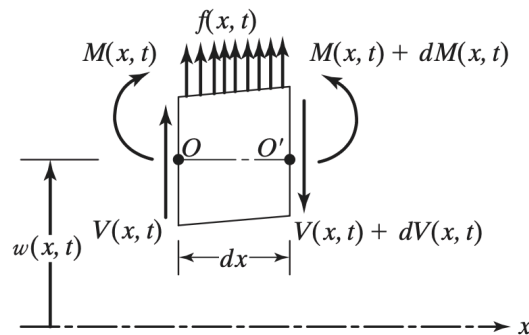


Fig 3.2 Free body diagram of an element of beam (credit: S.S. Rao, Mechanical Vibrations)

$M(x, t)$ is the bending moment, $V(x, t)$ is the shear force, and $f(x, t)$ is the external force per unit length of the beam.

3.2 FORMULATION OF GOVERNING EQUATION

From Newton's second law, the total force acting on element will be equal to its inertial force i.e.

$$-(V + dV) + f(x, t) dx + V = \rho A dx \frac{\partial^2 w}{\partial t^2}(x, t)$$

Moment equation about y-axis passing through O leads to

$$(M + dM) - (V + dV) dx + f(x, t) dx \cdot \frac{dx}{2} - M = 0$$

Simplifying using Taylor expansion, force equation and moment equation become

$$-\frac{\partial V}{\partial x}(x, t) + f(x, t) = \rho A(x) \frac{\partial^2 w}{\partial t^2}(x, t)$$

$$\frac{\partial M}{\partial x}(x, t) - V(x, t) = 0$$

Using $V = \frac{\partial M}{\partial x}$ from moment equation in force equation

$$-\frac{\partial^2 M}{\partial x^2}(x, t) + f(x, t) = \rho A(x) \frac{\partial^2 w}{\partial t^2}(x, t)$$

Moment-curvature relation (Euler–Bernoulli)

$$M(x, t) = EI(x) \frac{\partial^2 w}{\partial x^2}(x, t)$$

Substitute into equation to get equation of motion for forced lateral vibration of nonuniform beam:

$$\frac{\partial^2}{\partial x^2} \left[EI(x) \frac{\partial^2 w}{\partial x^2}(x, t) \right] + \rho A(x) \frac{\partial^2 w}{\partial t^2}(x, t) = f(x, t)$$

For a uniform beam, above equation simplifies to

$$EI \frac{\partial^4 w}{\partial x^4}(x, t) + \rho A \frac{\partial^2 w}{\partial t^2}(x, t) = f(x, t)$$

For free vibration (no external force), equation of motion becomes

$$c^2 \frac{\partial^4 w}{\partial x^4}(x, t) + \frac{\partial^2 w}{\partial t^2}(x, t) = 0 \quad (3.1)$$

where

$$c = \sqrt{\frac{EI}{\rho A}}$$

3.2.1 Initial conditions

Values of lateral displacement and velocity at $t = 0$,

$$w(x, t = 0) = w_0(x) \text{ and } \frac{\partial w}{\partial t}(x, t = 0) = \dot{w}_0(x)$$

3.2.2 Boundary Conditions

At Left End ($x = 0$) — Pinned Support:

$$\text{Displacement is zero: } w(0) = 0; \quad \text{Bending moment is zero: } \frac{\partial^2 w}{\partial x^2}(0) = 0$$

At Right End ($x = L$) — Vertical Sliding Support:

$$\text{Rotation is restricted (slope is zero): } \frac{\partial w}{\partial x}(L) = 0; \quad \text{Shear force is zero: } \frac{\partial^3 w}{\partial x^3}(L) = 0$$

3.2 NON-DIMENSIONALIZATION OF GOVERNING EQUATION

Divide the governing equation (3.1) by ρA :

$$\frac{EI}{\rho A} \frac{\partial^4 w}{\partial x^4}(x, t) + \frac{\partial^2 w}{\partial t^2}(x, t) = 0$$

Let $\xi = \frac{x}{L}$ and $\tau = t \sqrt{\frac{EI}{\rho A L^4}}$ be two dimensionless spatial and time variables. By chain rule,

$$\frac{\partial^4 w}{\partial x^4} = \frac{1}{L^4} \frac{\partial^4 w}{\partial \xi^4} \quad \text{and} \quad \frac{\partial^2 w}{\partial t^2} = \frac{EI}{\rho A L^4} \frac{\partial^2 w}{\partial \tau^2}$$

$$\text{Substituting into equation, we get } \frac{EI}{\rho A L^4} \frac{\partial^4 w}{\partial \xi^4} + \frac{\partial^2 w}{\partial \tau^2} = 0$$

Then final dimensionless equation becomes

$$\frac{\partial^4 w}{\partial \xi^4} + \frac{\partial^2 w}{\partial \tau^2} = 0$$

CHAPTER 4

ANALYTICAL SOLUTION

4.1 GENERAL SOLUTION OF BEAM EQUATION

By method of separation of variables

$$w(x, t) = W(x)T(t)$$

Substitute into governing equation and rearranging

$$\frac{c^2}{W(x)} \frac{d^4 W(x)}{dx^4} = - \frac{1}{T(t)} \frac{d^2 T(t)}{dt^2} = a = \omega^2$$

Above equation can be written as two equations:

$$\frac{d^4 W(x)}{dx^4} - \beta^4 W(x) = 0 \quad (\text{Spatial Ordinary Differential Equation})$$

$$\frac{d^2 T(t)}{dt^2} + \omega^2 T(t) = 0 \quad (\text{Temporal Ordinary Differential Equation})$$

Where

$$\beta^4 = \frac{\omega^2}{c^2} = \frac{\rho A \omega^2}{EI}$$

The solution of the Temporal Ordinary Differential Equation can be written as

$$T(t) = A \cos \omega t + B \sin \omega t$$

Where A and B are constants that can be found from initial conditions.

For the solution of the Spatial Ordinary Differential Equation, we assume the vibration is sinusoidal

$$W(x) = C e^{sx}$$

Where C and S are constants, and derive the auxiliary equation as

$$s^4 - \beta^4 = 0$$

The roots of this equation are $s_{1,2} = \pm \beta$ and $s_{3,4} = \pm i\beta$

Hence, the solution of the Spatial Ordinary Differential Equation can be written as

$$W(x) = C_1 \cos \beta x + C_2 \sin \beta x + C_3 \cosh \beta x + C_4 \sinh \beta x$$

The constants C_1, C_2, C_3, C_4 can be found from the boundary conditions. The natural frequency of the beam can be computed from

$$\omega = \beta^2 \sqrt{\frac{EI}{\rho A}} = (\beta l)^2 \sqrt{\frac{EI}{\rho A l^4}}$$

The function $W(x)$ is known as the normal mode or characteristic function of the beam and ω is called the natural frequency of vibration.

4.2 APPLYING BOUNDARY CONDITIONS

4.2.1 Boundary Condition 1 $\rightarrow W(0) = 0$

$$W(0) = 0 = C_1 \cos 0 + C_2 \sin 0 + C_3 \cosh(0) + C_4 \sinh(0)$$

$$C_1 + C_3 = 0$$

4.2.2 Boundary Condition 2 $\rightarrow W''(0) = 0$

$$W'(x) = -C_1 \beta \sin \beta x + C_2 \beta \cos \beta x + C_3 \beta \sinh \beta x + C_4 \beta \cosh \beta x$$

$$W''(x) = -C_1 \beta^2 \cos \beta x - C_2 \beta^2 \sin \beta x + C_3 \beta^2 \cosh \beta x + C_4 \beta^2 \sinh \beta x$$

Applying conditions at $x = 0$

$$-C_1 \beta^2 + C_3 \beta^2 = 0$$

$$C_1 - C_3 = 0$$

Therefore, solution at left end ($x = 0$) becomes

$$W(x) = C_2 \sin(\beta x) + C_4 \sinh(\beta x)$$

4.2.3 Boundary Condition 3 $\rightarrow W'(L) = 0$

$$W'(L) = -C_1\beta \sin(\beta L) + C_2\beta \cos(\beta L) + C_3\beta \sinh(\beta L) + C_4\beta \cosh(\beta L)$$

Since C_1 and C_3 are 0:

$$C_2 \cos(\beta L) + C_4 \cosh(\beta L) = 0$$

4.2.4 Boundary Condition 4 $\rightarrow W'''(L) = 0$

Using the third derivative at $x = L$:

$$-C_2\beta^3 \cos \beta L + C_4\beta^3 \cosh(\beta L) = 0$$

Simplifying,

$$-C_2 \cos \beta L + C_4 \cosh(\beta L) = 0$$

Solving the System of Equations

$$C_2 \cos(\beta L) + C_4 \cosh(\beta L) = 0 \quad -C_2 \cos(\beta L) + C_4 \cosh(\beta L) = 0$$

$$2C_4 \cosh(\beta L) = 0$$

Since $\cosh(\beta L)$ can never be 0. So, $C_4 = 0$

For a Non-Trivial Solution ($C_2 \neq 0$) : $\cos(\beta L) = 0$

Therefore, the eigenvalues are:

$$\beta L = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2} \dots$$

Therefore, the natural frequency ω_n is:

$$\omega_n = (\beta_n L)^2 \sqrt{\frac{EI}{\rho A L^4}}$$

And the mode shape $W(x)$ is:

$$W(x) = C_2 \sin(\beta x)$$

For normalization, $C_2 = 1$

$$W(x) = \sin(\beta x)$$

4.3 CALCULATING FIRST THREE NATURAL FREQUENCIES AND CORRESPONDING MODE SHAPES

4.3.1 Beam Properties and Parameters

- Length, $L = 500 \text{ mm} = 0.5 \text{ m}$
- Cross-sectional dimensions, $b = 20 \text{ mm}$, $h = 20 \text{ mm}$
- Cross-sectional area, $A = b \times h = 20 \times 20 = 400 \text{ mm}^2 = 4 \times 10^{-4} \text{ m}^2$
- Second Moment of Area, $I = \frac{bh^3}{12} = \frac{20 \times 20^3}{12} = 13333.33 \text{ mm}^4 = 1.3333 \times 10^{-8} \text{ m}^4$
- Young's Modulus, $E = 210 \text{ GPa} = 2.1 \times 10^{11} \text{ Pa}$
- Density, $\rho = 7.85 \times 10^{-6} \text{ kg/mm}^3 = 7850 \text{ kg/m}^3$

4.3.2 Calculating Natural Frequencies

Using $\omega_n = \beta_n^2 \sqrt{\frac{EI}{\rho A}}$

First Natural Frequency ($n = 1$): $\omega_1 = \left(\frac{\pi}{2L}\right)^2 \sqrt{\frac{EI}{\rho A}}$

Second Natural Frequency ($n = 2$): $\omega_2 = \left(\frac{3\pi}{2L}\right)^2 \sqrt{\frac{EI}{\rho A}}$

Third Natural Frequency ($n = 3$): $\omega_3 = \left(\frac{5\pi}{2L}\right)^2 \sqrt{\frac{EI}{\rho A}}$

Plugging in the material properties for the beam,

$$\sqrt{\frac{EI}{\rho A}} = 29.14830583197063$$

$$\omega_n = \beta_n^2 \cdot 29.14830583197063$$

ω_1 :

$$\beta_1^2 = 9.869604401089358$$

$$\omega_1 = 287.7074068105146 \text{ rad/s}$$

ω_2 :

$$\beta_2^2 = 88.82643960980423$$

$$\omega_2 = 2589.366661294631 \text{ rad/s}$$

ω_3 :

$$\beta_3^2 = 246.74011002723395$$

$$\omega_3 = 7192.685170262865 \text{ rad/s}$$

Converting to Hertz:

$$f = \frac{\omega}{2\pi}$$

- $f_1 = 45.78 \text{ Hz}$

- $f_2 = 412.01 \text{ Hz}$

- $f_3 = 1144.92 \text{ Hz}$

4.3.3 Calculating Mode Shapes

$$W_n(x) = \sin(\beta_n x)$$

- $W_1(x) = \sin(\pi x)$
- $W_2(x) = \sin(3\pi x)$
- $W_3(x) = \sin(5\pi x)$

4.4 PLOTTING NORMALIZED MODE SHAPES

The first three mode shapes were plotted using MATLAB (see Appendix for MATLAB code). The analytical expression $W_n(x) = \sin(\beta_n x)$ was used. The mode shapes were normalized by dividing each function by its maximum absolute value. The spatial domain was taken from $x = 0$ to $x = L$.

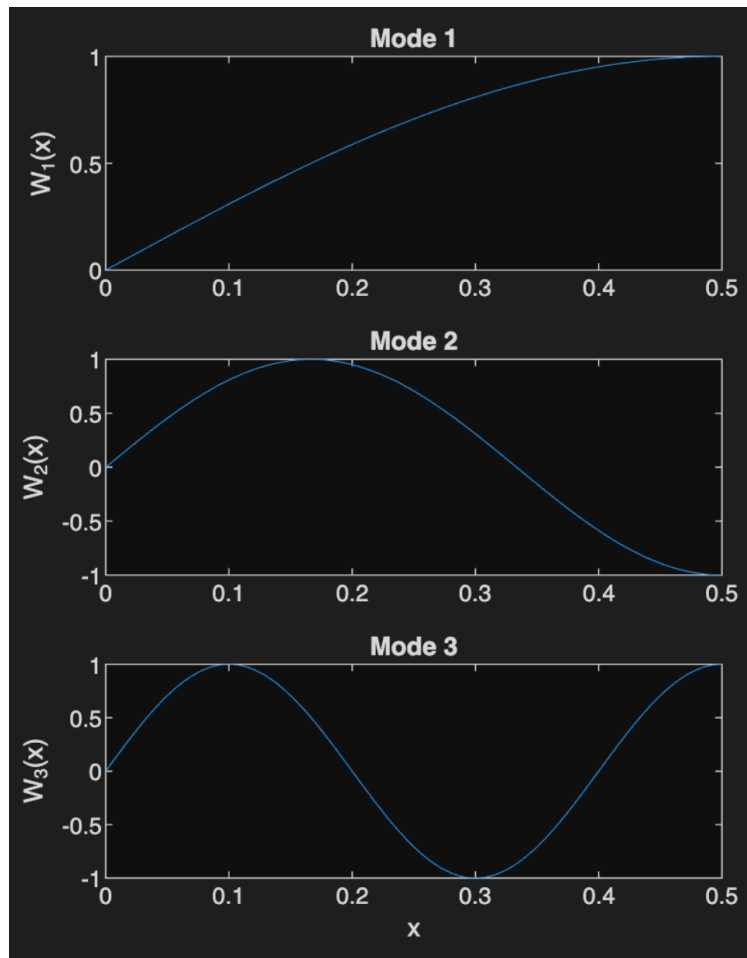


Fig 4.4.1 First three normalized mode shapes of the beam

Observations:

- Mode 1 shows a single half-wave with no internal nodes
- Mode 2 shows one internal node
- Mode 3 shows two internal nodes
- All mode shapes satisfy the boundary conditions
- The number of nodes increases with mode number

4.5 EFFECT OF THE SLIDING BOUNDARY ON MODE SHAPE CHARACTERISTICS

The sliding support on the side of the beam stops it from rotating. This means the beam is completely flat at that point. If you look at all the mode shapes you will see the beam ends with a tangent, which is really clear.

With this limit the mode shapes are still really smooth and do not have any sudden changes near the edge. The main thing the sliding support does is not change the shape of the beam. It does change the wavelengths of vibration that are allowed.

So the mode shapes follow a pattern of the beam the mode shapes follow a pattern of $(2n - 1)\frac{\pi}{2}$ of the usual $n\pi$ seen in simply supported beams the mode shapes of the beam. This affects the frequencies of the beam and makes sure the slope condition, at the sliding end of the beam is satisfied.

In summary the sliding boundary of the beam mainly affects the behaviour of the beam, which is the zero slope and the frequency values of the beam while the overall sinusoidal nature of the mode shapes of the beam remains the same.

CHAPTER 5

FEM ANALYSIS

5.1 MODEL SETUP

A 3D rectangular beam model of length 0.5 m and cross sectional area of 20x20 mm² was created in ANSYS.

Structural steel was selected due to its well-defined mechanical properties, it is commonly used in structural applications and provides reliable results for modal analysis. Cross-section was assumed to be uniform and modal analysis was performed to extract natural frequencies.

For Body Sizing, an element size of 5 mm was selected to ensure sufficient mesh refinement for accurate capture of deformation while maintaining reasonable computational cost.

5.2 BOUNDARY CONDITIONS

1. Left End (Pin Support):

ANSYS Setup:

- Remote Displacement Translations (X, Y, Z) = 0
- Rotation Z = Free: Allows the face to pivot for bending. Rotation X, Y = 0: Stops unrealistic 3D twisting.

2. Right End (Vertical Sliding Support):

ANSYS Setup:

- Rotations (X, Y, Z) = 0: Forces the face to stay perfectly vertical while sliding up and down.
- Remote Displacement (Right Face) Translation Y = Free: Allows vertical vibration. Translation X, Z = 0: Stops horizontal sliding

In ANSYS, the sliding condition was approximated using displacement constraints, which may not perfectly enforce theoretical shear-free conditions.

5.3 NATURAL FREQUENCIES

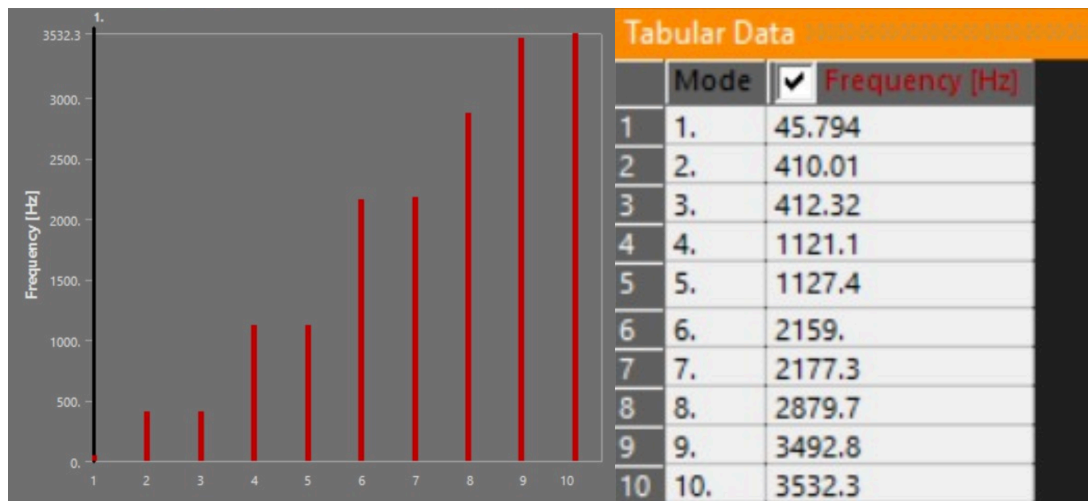


Fig 5.3.1 Natural Frequencies obtained from FEM analysis

Frequencies 1, 2 and 5 correspond to first three modes calculated in section 4.3.3 using Euler-Bernoulli Analytical Model i.e. corresponding to transverse vibration only but rest of the modes in the above figure correspond to a combination of transverse, lateral and twisting deformations.

5.4 MODE SHAPES

Mode 1 shows a smooth deformation with maximum displacement at the right end.

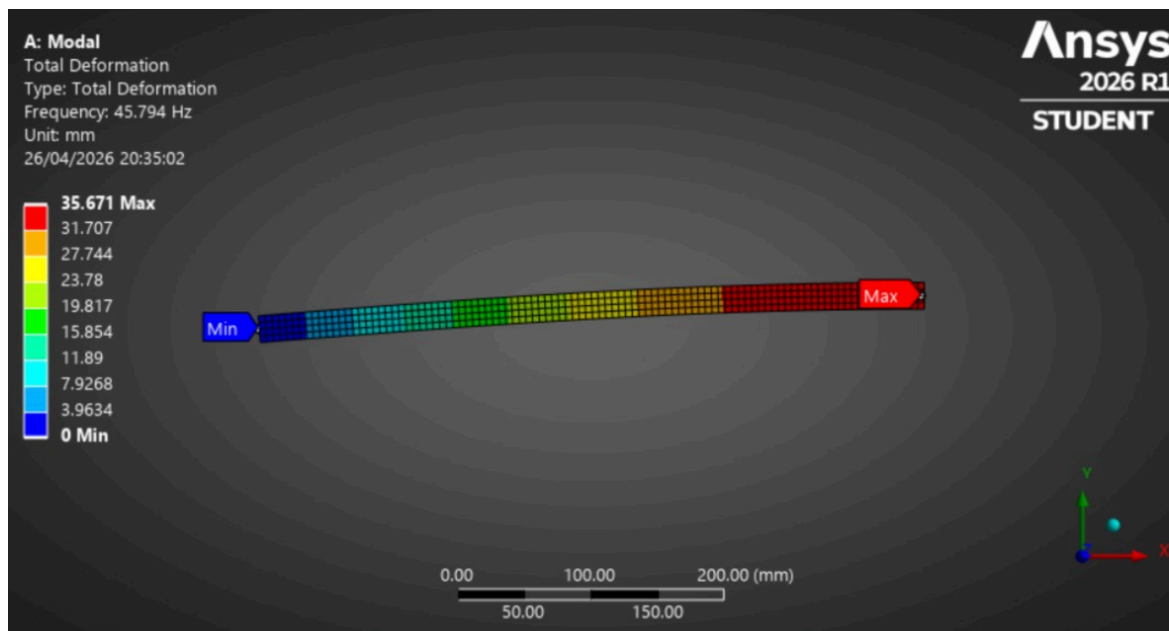


Fig 5.4.1 Mode Shape 1 (45.794 Hz)

Mode 2 introduces one node along the length.

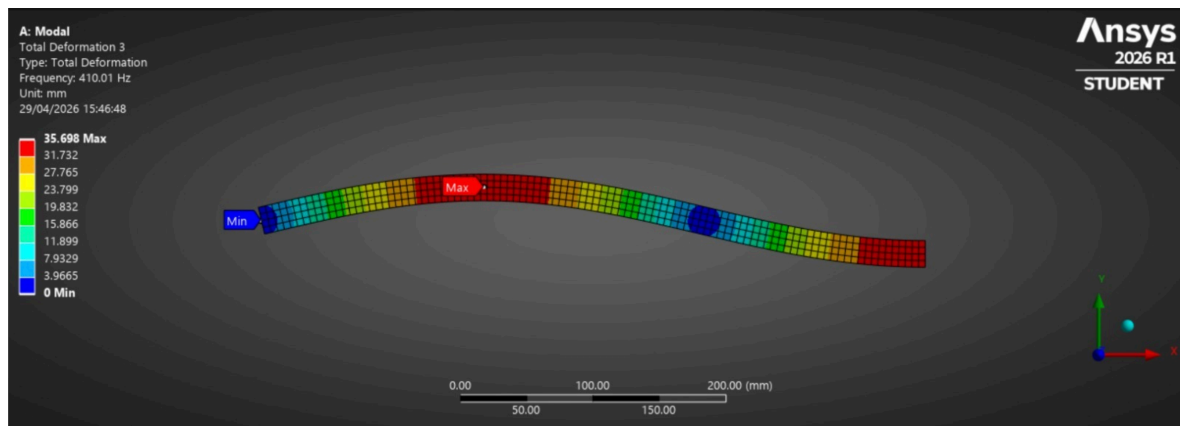


Fig 5.4.2 Mode Shape 2 (410.01 Hz)

Mode 3 shows two nodes, indicating higher vibration mode.

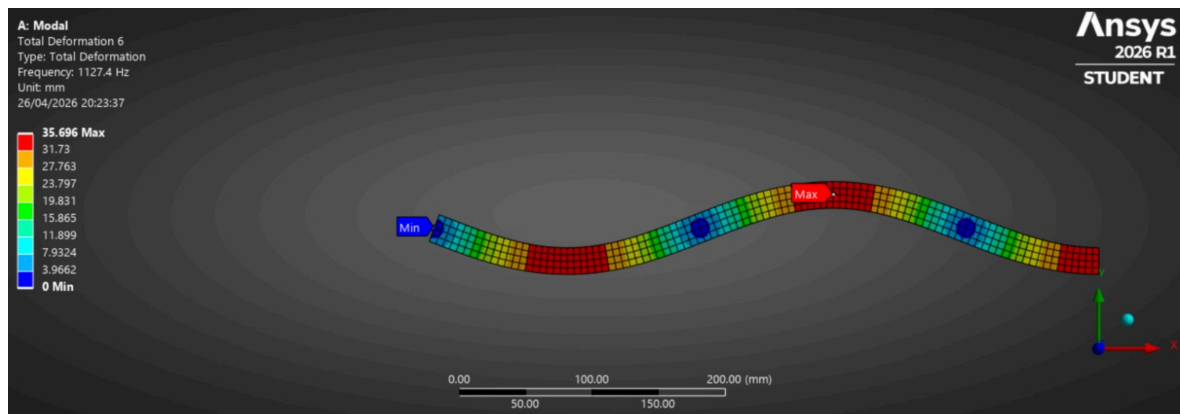


Fig 5.4.3 Mode Shape 3 (1127.4 Hz)

5.5 COMPARISON WITH ANALYTICAL RESULTS

Mode	Analytical (Hz)	FEM (Hz)	Error (%)
1	45.78	45.794	0.03%
2	412.01	410.01	0.49%
3	1144.92	1127.3	1.54%

Table 5.1 Comparison of Analytical Results with FEM analysis results

5.6 DISCUSSION

The strong correlation between the analytical Euler-Bernoulli results and the FEM (ANSYS) data, which shows a maximum error of only 1.54%, confirms the model. However, the growing difference in higher modes calls for a technical review:

- **Boundary Condition Idealisation:** The analytical approach assumes a "perfect" sliding support with no friction and no shear force. In the FEM environment, the constraints used to simulate this slider often add a small amount of artificial stiffness. This prevents the end from being truly "force-free," causing the slight frequency shifts shown in Table 5.5.1.
- **Theoretical Simplifications:** Euler-Bernoulli theory treats the beam as a 1D line, ignoring the physical thickness of the material. As the mode number increases and the vibration wavelength shortens, 3D effects like internal shear and rotary inertia become more important. Since the FEM model considers the complete 3D volume, it captures extra flexibility that the 1D math cannot account for, resulting in a larger gap in the higher harmonics.
- **Parasitic 3D Vibrations:** In a 3D simulation, energy can be lost to small lateral wobbles or twisting motions that are not included in the 1D vertical equations. These secondary motions slightly lower the primary frequencies in the simulation, especially in the more complex third mode.
- **Mesh Discretisation:** Although the 0.03% error in Mode 1 indicates a high-quality mesh, the process of dividing a smooth beam into finite "blocks" introduces small numerical errors. These discrepancies naturally build up as the vibration patterns become more complex in higher modes.

References

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- Rao, S. S., & Griffin, P. (2018). *Mechanical vibrations*. Harlow Pearson.
- Reddy, J. N. (2018). *Introduction to the Finite Element method*. McGraw-Hill.

APPENDIX

MATLAB CODE FOR PLOTTING MODE SHAPES

```
L = 0.5;
x = linspace(0,L,1000);
beta1 = pi;
beta2 = 3*pi;
beta3 = 5*pi;
w1 = sin(beta1*x);
w2 = sin(beta2*x);
w3 = sin(beta3*x);
w1 = w1./max(abs(w1));
w2 = w2./max(abs(w2));
w3 = w3./max(abs(w3));
subplot(3,1,1)
plot(x,w1)
ylabel('W_1(x)')
title('Mode 1')
subplot(3,1,2)
plot(x,w2)
ylabel('W_2(x)')
title('Mode 2')
subplot(3,1,3)
plot(x,w3)
xlabel('x')
ylabel('W_3(x)')
title('Mode 3')
```